

Circuit Simulation Lab:8

Demultiplexers

Introduction

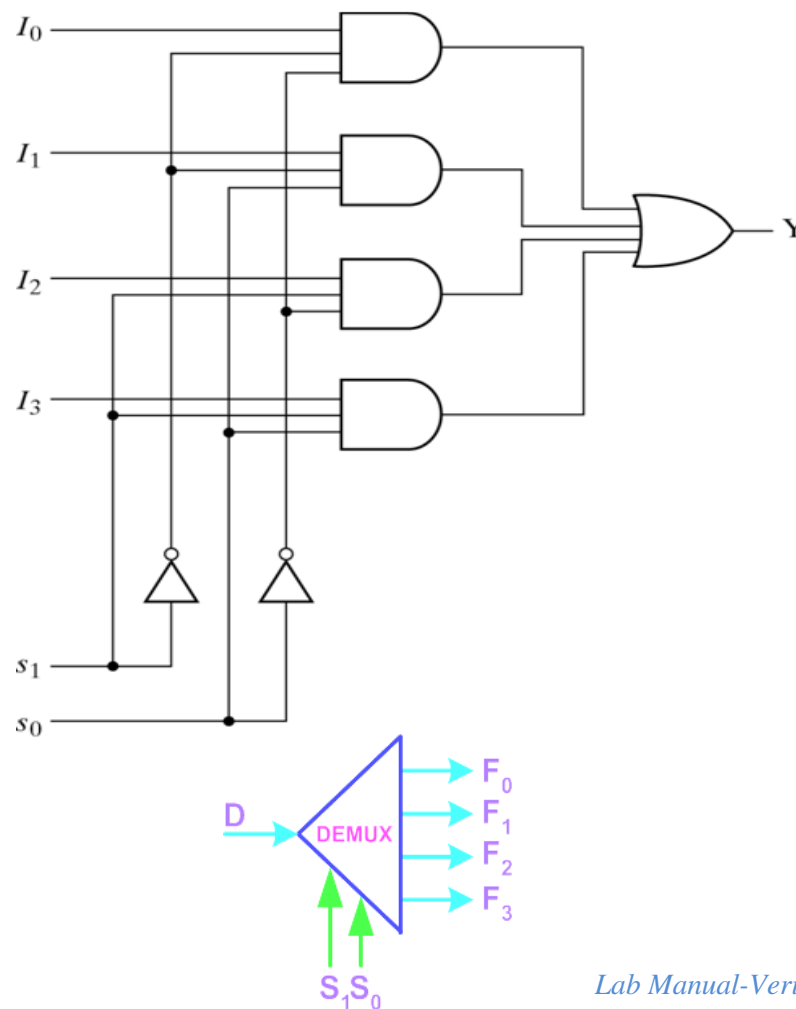
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Demultiplexer. Demultiplexer or Demux is a combinational circuit that distributes the single input data to a specific output line. The control inputs or selection lines are used to select a specific output line from the possible output lines.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



1:4 Demultiplexer

S1	S0	F0	F1	F2	F3
0	0	0	0	0	0
0	1	0	0	0	0
1	0	0	0	0	0
1	1	0	0	0	0

Truth Table

Describe the above in Verilog HDL and capture the Waveforms

1. *Dataflow modeling*
2. *Behavior modeling*
3. *Structural modeling (Use the gates for designing the Mux)*
4. *Design a 1:8 Demux using two 1:4 Demux(Structural Modeling)*